Solution

Matrix Theory, Test 1N February 4, 2026

(1) (25 pts.) Write the augmented matrix for the system of linear equations:

$$\begin{aligned} 2x - y \quad &= 3 \\ x - y \quad &= -2 \\ 3x - 2y + z &= -1 \end{aligned}$$

For the following, A always denotes the matrix of coefficients, $\mathbf{b}$ the right hand side vector, and $[A \mid \mathbf{b}]$ the augmented matrix.

15 (a) Use elementary row operations to reduce the augmented matrix $[A \mid \mathbf{b}]$ into reduced row echelon form.

4 (b) Write the solution to the original system of equations.

3@2 (c) True or False? The original system of linear equations is consistent.

(d) True or False? The columns of A are linearly independent vectors.

(e) True or False? The columns of the matrix $[A \mid \mathbf{b}]$ are linearly independent vectors.

a. $\left[\begin{array}{ccc|c} 2 & -1 & 0 & 3 \\ 1 & -1 & 0 & -2 \\ 3 & -2 & 1 & -1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & -1 & 0 & -2 \\ 2 & -1 & 0 & 3 \\ 3 & -2 & 1 & -1 \end{array}\right] \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -1 & 0 & -2 \\ 0 & 1 & 0 & 7 \\ 0 & 1 & 1 & 5 \end{array}\right]$

$\xrightarrow[R_3 - R_2]{R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -2 \end{array}\right]$

b. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ -2 \end{pmatrix}$

c. True

d. True

e. False

Version 1M

a. $\left[\begin{array}{ccc|c} 2 & -1 & 0 & 1 \\ 1 & -1 & 0 & -3 \\ 3 & -2 & 1 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 2 & -1 & 0 & 1 \\ 3 & -2 & 1 & 1 \end{array}\right] \times -3$

$\to \left[\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 0 & 1 & 0 & 7 \\ 0 & 1 & 1 & 10 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 3 \end{array}\right]$

b. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \\ 3 \end{pmatrix}$

c. T

d. T

e. F

(2) (25 pts.)

$$A = \begin{bmatrix} 1 & 3 & 2 \\ -1 & -2 & 1 \\ 0 & 1 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 11 \end{bmatrix}$$

13 (a) Using elementary row operations, find the reduced row echelon form of the augmented matrix $[A \mid \mathbf{b}]$.

4 (b) Describe all solutions to $A\mathbf{x} = \mathbf{0}$ in parametric vector form.

4@2
(c) True or False? The system of linear equations $A\mathbf{x} = \mathbf{b}$ is consistent.
(d) True or False? The columns of A are linearly independent vectors.
(e) True or False? The vector $\mathbf{b}$ is in the range of the linear transformation $T(\mathbf{x}) = A\mathbf{x}$.
(f) True or False? The equation $A\mathbf{x} = \mathbf{y}$ has a solution for each $\mathbf{y}$ in $\mathbb{R}^3$.

a. $\left[\begin{array}{ccc|c} 1 & 3 & 2 & 2 \\ -1 & -2 & 1 & 1 \\ 0 & 1 & 3 & 11 \end{array}\right] \xrightarrow{R_2+R_1} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 1 & 3 & 11 \end{array}\right] \xrightarrow[R_3-R_2]{R_1-3R_2} \left[\begin{array}{ccc|c} 1 & 0 & -7 & -7 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 0 & 8 \end{array}\right]$

$\left[\begin{array}{ccc|c} 1 & 0 & -7 & -7 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_2-3R_3]{R_1+7R_3} \left[\begin{array}{ccc|c} 1 & 0 & -7 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$

$\frac{1}{8}R_3$

b. $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7x_3 \\ -3x_3 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}$

c. False

d. False

e. False

f. False

Version 1M

a. $\left[\begin{array}{ccc|c} 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 11 \\ -1 & -2 & 1 & 1 \end{array}\right]$

$\rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 11 \\ 0 & 1 & 3 & 3 \end{array}\right] \xrightarrow{-3} \left[\begin{array}{ccc|c} 1 & 0 & -7 & -29 \\ 0 & 1 & 3 & 11 \\ 0 & 0 & 0 & -8 \end{array}\right]$

$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -7 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$

b. $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \begin{bmatrix} 7 \\ -3 \\ 1 \end{bmatrix}$

c. F

d. F

e. F

f. F

(3) (25 pts.)

$$A = \begin{bmatrix} -1 & 2 & 1 & 0 \\ -2 & 4 & 2 & 1 \\ 2 & -4 & -1 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

12 (a) Using elementary row operations, find the reduced row echelon form of the augmented matrix $[A \mid \mathbf{b}]$.

7 (b) Describe the general solution to the matrix equation $A\mathbf{x} = \mathbf{b}$ in parametric vector form.

3@2 (c) True or False? The columns of A are linearly independent vectors.

(d) True or False? The vector $\mathbf{b}$ is in the range of the linear transformation $T(\mathbf{x}) = A\mathbf{x}$.

(e) True or False? The equation $A\mathbf{x} = \mathbf{y}$ has a solution for each $\mathbf{y}$ in $\mathbb{R}^3$.

a. $\left[\begin{array}{cccc|c} -1 & 2 & 1 & 0 & 2 \\ -2 & 4 & 2 & 1 & 1 \\ 2 & -4 & -1 & -1 & 1 \end{array}\right] \begin{array}{l} +2 \\ \\ \\ \end{array} \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 + R_1}]{-R_1} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 1 & -1 & 5 \end{array}\right] \xrightarrow[R_2 \leftrightarrow R_3]{R_1 + R_3} \left[\begin{array}{cccc|c} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & -1 & 5 \\ 0 & 0 & 0 & 1 & -3 \end{array}\right]$

$\xrightarrow[R_2 + R_3]{R_1 + R_3} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{array}\right]$

b. $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \\ -3 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

c. False
d. True
e. True

Version 1M

a. $\left[\begin{array}{cccc|c} -1 & 2 & 1 & 0 & 2 \\ 2 & -4 & -1 & -1 & 1 \\ -2 & 4 & 2 & 1 & 1 \end{array}\right] \begin{array}{l} -2 \\ \\ \\ \end{array} \longrightarrow \left[\begin{array}{cccc|c} 1 & -2 & -1 & 0 & -2 \\ 0 & 0 & 1 & -1 & 5 \\ 0 & 0 & 0 & 1 & -3 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|c} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & -1 & 5 \\ 0 & 0 & 0 & 1 & -3 \end{array}\right]$

$\longrightarrow \left[\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{array}\right]$

b. $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \\ -3 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

(4) (25 pts.) Write the augmented matrix for the system of linear equations:

$$-x + y - z = -1$$
$$2x - 3y + z = -2$$

For the following, A always denotes the matrix of coefficients, $\mathbf{b}$ the right hand side vector, and $[A \mid \mathbf{b}]$ the augmented matrix.

10 (a) Use elementary row operations to reduce the augmented matrix $[A \mid \mathbf{b}]$ into reduced row echelon form.

7 (b) Write the solution to the original system of equations in parametric vector form.

(c) True or False? The original system of linear equations is consistent.

4@2 (d) True or False? The columns of A are linearly independent vectors.

(e) True or False? The vector $\mathbf{b}$ is in the range of the linear transformation $T(\mathbf{x}) = A\mathbf{x}$.

(f) True or False? The equation $A\mathbf{x} = \mathbf{y}$ has a solution for each $\mathbf{y}$ in $\mathbb{R}^2$.

a. $\left[\begin{array}{ccc|c} -1 & 1 & -1 & -1 \\ 2 & -3 & 1 & -2 \end{array}\right]^2 \xrightarrow[R_2+2R_1]{-R_1} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & -1 & -1 & -4 \end{array}\right] \begin{array}{c} R_1-R_2 \\ -R_2 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 5 \\ 0 & 1 & 1 & 4 \end{array}\right]$

b. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5-2z \\ 4-z \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} + z\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

c. T d. F e. T f. T

Version 1M

a. $\left[\begin{array}{ccc|c} -1 & 1 & -1 & 1 \\ 3 & -2 & 4 & 2 \end{array}\right]^3 \longrightarrow \left[\begin{array}{ccc|c} 1 & -1 & 1 & -1 \\ 0 & 1 & 1 & 5 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & 1 & 5 \end{array}\right]$

b. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4-2z \\ 5-z \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} + z\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

c. T d. F e. T f. T